التاريخ: 2025/2/6

الموضوع: عباس اسعد / محمد المهدي عبد الحميد حميد

$x(t) = \frac{1}{2\pi}\cos(4000\pi t)\cos(1000\pi t)$

* $\cos x \cos y = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$ *

$x(t) = \frac{1}{2\pi}\cdot\frac{1}{2}\left[\cos 5000\pi t + \cos 3000\pi t\right]$

$f_1 = \frac{5000\pi}{2\pi} = 2500\,Hz$

$f_2 = \frac{3000\pi}{2\pi} = 1500\,Hz$

$\rightarrow f_{max} = 2.5\,KHz$

$\boxed{F_s = 2f_{max} = 5\,KHz}$ $\rightarrow$ Nyquist rate

$\boxed{T_s = 1/F_s = 2\times10^{-4}\,s}$ $\rightarrow$ Nyquist interval

التاريخ: / /

الموضوع:

6.1.1 تكملة من سؤال

B.W of $g_1(t) g_2(t)$ is the summation of B.W of $g_1(t)$ and $g_2(t)$

$$B.W = 100KHZ + 150KHZ = \boxed{250KHZ}$$

Nyquist sample rate $= 2B = 2 * 250 KHZ$

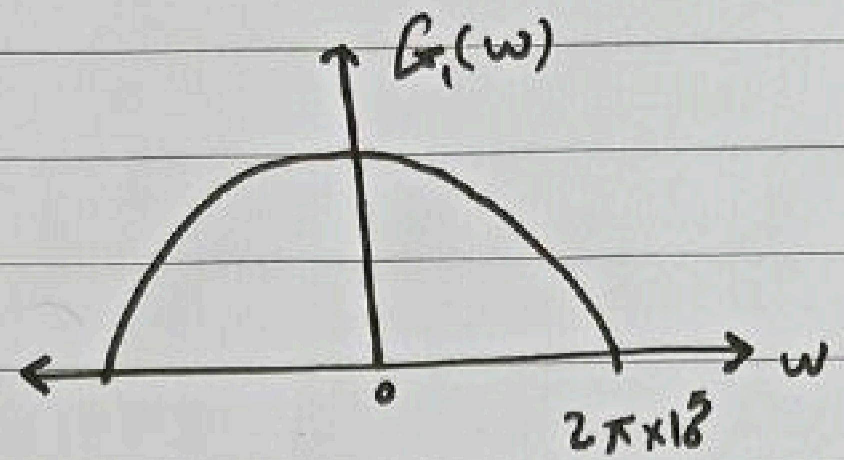
$$f_s = 500KHZ$$

Nyquist interval $= \frac{1}{f_s} = 2 \mu sec$

التاريخ: / /   الموضوع:

Problem 6.11

Fig. shows Fourier spactra of signals $g_1(t)$ & $g_2(t)$.
Determine the Nyquist interval & the samp rate for signal $g_1(t)$, $g_2(t)$, $g_1^2(t)$ & $g_1(t)*g_2(t)$



Sol:; B.W $g_1(t) = 2\pi * 100000 = w = 2\pi f$

$\therefore f = \frac{\cancel{2\pi} * 100{,}000}{\cancel{2\pi}} = \boxed{100k\ HZ}$

Nyqist sample rate f_s for $g_1(t) = 2B = 2 * 100k = \boxed{200kHZ}$

Nyquist interval of $g_1(t) = \frac{1}{f_s} = \frac{1}{200k} = \boxed{5\mu sec}$

B.W of $g_2(t) = \boxed{150\ kHZ}$ for $g_1(t) = 2B = 2*150k = \boxed{300kHZ}$ Nyquist sample rate

Nyquist interval of $g_2(t) = \frac{1}{f_s} = \frac{1}{300kHZ} = \boxed{3\ \mu sec}$

B.W of $g_1^2(t)$ is twice of $g_1(t)$ $2*100k\ HZ = \boxed{200\ KHZ}$

Nyquist sample rate of $g_1^2(t) \Rightarrow f_s = 2B = \boxed{400\ KHZ}$

Nyquist interval $= \frac{1}{f_s} = \frac{1}{400KHZ} = \boxed{2.5\ \mu sec}$

6.2-4

Each signal has B.W = 1.2KHZ

signal كل حسب السؤال

$\overset{2}{2}40HZ * 5 = 1200HZ = \boxed{1.2KHZ}$

عدد Signals

$$\nabla V = \frac{2mp}{L}$$

Quantization error ≤ 20% * mp * Nyqiast rate

6.2-5 / B.W = 100HZ

Nyquist rate = 2 * B.W = 2 * 100 = 200 HZ

The sample rate = 2 * Nyquist rate = 2 * 200 = $\underbrace{400HZ}$ for 1 signal

I have (10 patients) = 10 samples
حسب السؤال

For 10 patients = 10 * 400 = $\boxed{4000}$ $\boxed{\text{sample * HZ}}$ or $\boxed{\frac{\text{sample}}{\text{seconds}}}$

ΔV is quatization step T.S = $\frac{1}{400}$ =

Muhammed Al-Mahdi | ٣ نيسان ٢٠٢٥ في ٢٣:٢٨

~~c.co/c/c~~

c.co/c/c. Communication

الخميس

c. Laith

6.2-3 P.323

BW = 4.5 MHZ

rate = 20%

(a) sampling rate

$$\text{sampling rate} = \left[9 + \frac{20}{100} * 9\right] = [9+1.8] = \boxed{10.8\ \text{MHZ}}$$

(b) L = 1024

$$L = 2^n \Rightarrow 2^n = 1024 \quad \therefore \boxed{n = 10}$$

(c)

$$\boxed{10.8 * 10^6} * \boxed{10} = 108\ \text{Mbit/sec}$$

٢٠٢٥/٣/١٣ Communication System III

الخميس T. [redacted] L. Laith

$$C = \begin{cases} \dfrac{3\,\widetilde{m^2(t)}}{m_p^2} & \text{uncompresed case} \\[2em] \dfrac{3L^2}{[\ln(1+\mu)]^2} & \text{compresed case} \end{cases}$$

جداً مهم

6.2-10 Given no. of bit $n = 10$ $SNR = 30\,dB$

We know that SNR in PCM is:-

$$\left(\frac{S_o}{N_o}\right)_{dB} = (\alpha + 6n)_{dB} \quad \ldots (1)$$ $\boxed{\text{in Pag } 279}$ ★★

where $\alpha = 10 \log_{10}(C)$

Thus increasing (n) by (1 bit) → increase $\left(\frac{S_o}{N_o}\right)$ by (6 dB)

Now, $SNR = 30\,dB$ → desired $SNR = 42\,dB$

For 12 dB incerase, (n) has increas by (2) bit $\boxed{n = 10 + 2 = 12}$

Fraction increas in B.W

$$= \frac{12 - 10}{10} * 100\% = 20\%$$

~~B.W~~ $\boxed{\text{Increase of B.W is } 20\%}$

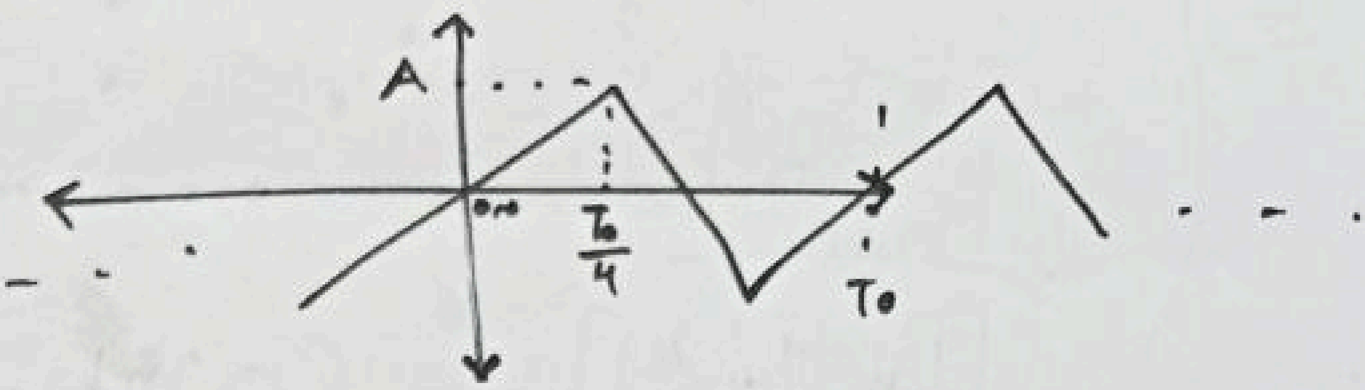
6.2.7

Let the peak Amp (A), then (T_0) is the period of $m(t)$ from the question each quarter cycles on the same Amp. value. Therefore each quartor cycle contribution identical energy, so the power ~~the~~ of this signal can be computed by eq. of firest quarter cycle.

$$m(t) = 4\frac{A}{T_0}t$$

* $y = mx + b$

من معادلة الخط المستقيم



Power of mean square value (Energy Average over quarter cycle) is

$$\overline{m^2(t)} = \frac{1}{\frac{T_0}{4}}\int_0^{\frac{T_0}{4}} \left(\frac{4At}{T_0}\right)^2 dt \Rightarrow = \frac{4}{T_0} * \frac{16A^2}{T_0^2}\int_0^{T_0/4} t^2\, dt$$

$$\boxed{\overline{m^2(t)} = \frac{A^2}{3}}$$

$$\frac{\overline{m^2(t)}}{m_p^2} = \frac{\frac{\cancel{A^2}}{3}}{\cancel{A^2}} = \boxed{\frac{1}{3}}$$

Now, $SNR = 3L^2 \frac{\overline{m^2(t)}}{m_p^2}$

$$50119 = \cancel{3}L^2 * \boxed{\frac{1}{\cancel{3}}}$$

$\therefore L^2 = 50119$

$$\boxed{L = 223.872}$$

☆ From Previous prob. SNR = 47 dB

In Linear $\Rightarrow 47 = 10\log_{10}(SNR) \Rightarrow 4.7 = \log_{10}(SNR)$

$$\boxed{SNR = 10^{4.7} = 50119}$$

$$n = \lceil \log_2(L) \rceil = \log_2(223.872)$$

$$\Rightarrow \boxed{n = 7.8 \approx 8}$$

$L = 2^n = 2^8 = 256$ min. value of n required

$$\frac{S_o}{N_o} = 3L^2 \frac{\overline{m^2(t)}}{m_p^2} = 3(256)^2\frac{1}{3} = \boxed{\begin{array}{l}65536\\48.165\,dB\end{array}}$$

Q.2.8 | compression per matter $\mu = 100$

$$SNR = 45\,dB = 10\log_{10}(SNR) = 10^{4.5} = 31622\ \cancel{777}$$

we know that this SNR with compression we use the:

$$SNR = \frac{3L^2}{[\ln(1+\mu)]^2} \Rightarrow 31622777 = \frac{3L^2}{[\ln(101)]^2} \Rightarrow \therefore \boxed{L = 473.829}$$

$$\boxed{n = \lceil \log_2(L) \rceil = 8.888 \approx 9}$$

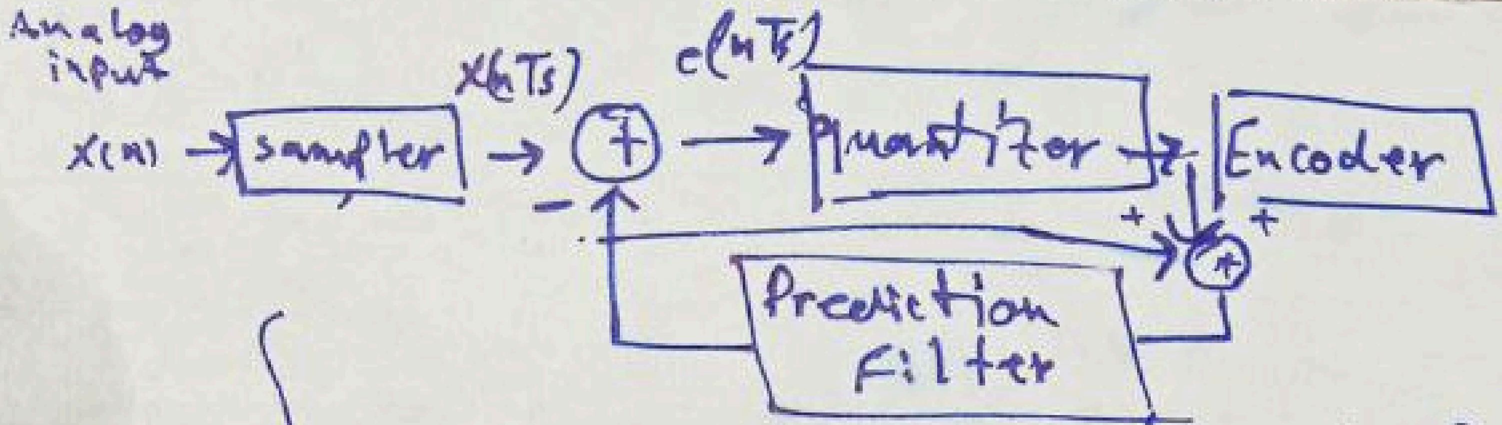
$\therefore L = 2^9 = 512$

Then the SNR for this (L)

$$SNR = \frac{3L^2}{[\ln(101)]^2}$$

$$\boxed{\begin{aligned} SNR &= 36923 \\ SNR &= 45.67\,dB \end{aligned}}$$

٢٠٢٥/٣/٢٠
الخميس



Ex:

Let input sampls $x(n) = \{2.1, 2.2, 2.3, 2.6, 2.7, 2.8\}$

Appling DPCM what is the Encoder & Decoder is done

Let the first order prediction filter $x_q(n-1)$?

Sol:

~~$\hat{x}(n) = x_q(n-1)$ $\quad e(n) = x(n) - \hat{x}(n)$~~

$\boxed{\text{Encoder}}$

$x(n)$	$\hat{x}(n) = x_q(n-1)$	$e(n) = x(n) - \hat{x}(n)$	$e_q(n)$	$\hat{x}_q(n) = \hat{x}(n) + e_q(n)$
2.1	0	2.1 - 0 = 2.1	2	0+2 = 2
2.2	2	2.2 - 2 = 0.2	0	2+0 = 2
2.3	2	2.3 - 2 = 0.3	0	2+0 = 2
2.6	2	2.6 - 2 = 0.6	1	2+1 = 3
2.7	3	2.7 - 3 = -0.3	0	3+0 = 3
2.8	3	2.8 - 3 = -0.2	0	3+0 = 3

Transmitted sequence:

$\{\underset{010}{2}\ \underset{000}{0}\ \underset{000}{0}\ \underset{001}{1}\ \underset{000}{0}\ \underset{000}{0}\}$ DPCM

bit=3

6.2.9 2*

A

signal band ~~limited~~ width limited at 1 MHz 10^6 Hz.

We know that the Nyquist rate twise the B-W.

Nyquist rate $= 2 * 10^6$ Hz.

signal is sampled at 50% above (Higher) the Nyquist rate.

$$\therefore \text{Sample rate} = \Big[\underbrace{2*10^6}_{N\text{-}R} + \underbrace{\frac{50}{100} * 2 * 10^6}_{50\% \text{ Higher or above than } N\text{-}R}\Big] = 3 * 10^6 \text{ Hz}$$

$$\boxed{L = 256 \;,\; \mu = 255 \quad = \text{Lbe}}$$

$\therefore$ signal to quantization noise Ratio $= \dfrac{3L^2}{[\ln(1+\mu)]^2}$

$$\frac{S_o}{N_o} = \frac{3(256)}{(\ln(256))^2} = \boxed{6394.19}$$

6.2.9 B

If the sampling is reduced, and the value of L is increased ~~→~~ then same no. of bit/s is maintained.

SNR can be improved with the same B.W.

$L = 256 \rightarrow n = 8 \rightarrow$ Transmission rate: ~~$3*10^6*8$~~

انتبه ☆☆

☆Transmission rate = sampling rate * n $= \boxed{3 * 10^6 * 8}$

New sample rate $= [2*10^6 + \frac{20}{100} * 2 * 10^6] = 2.4 * 10^6$ Hz

20% higher than Nyquist rate

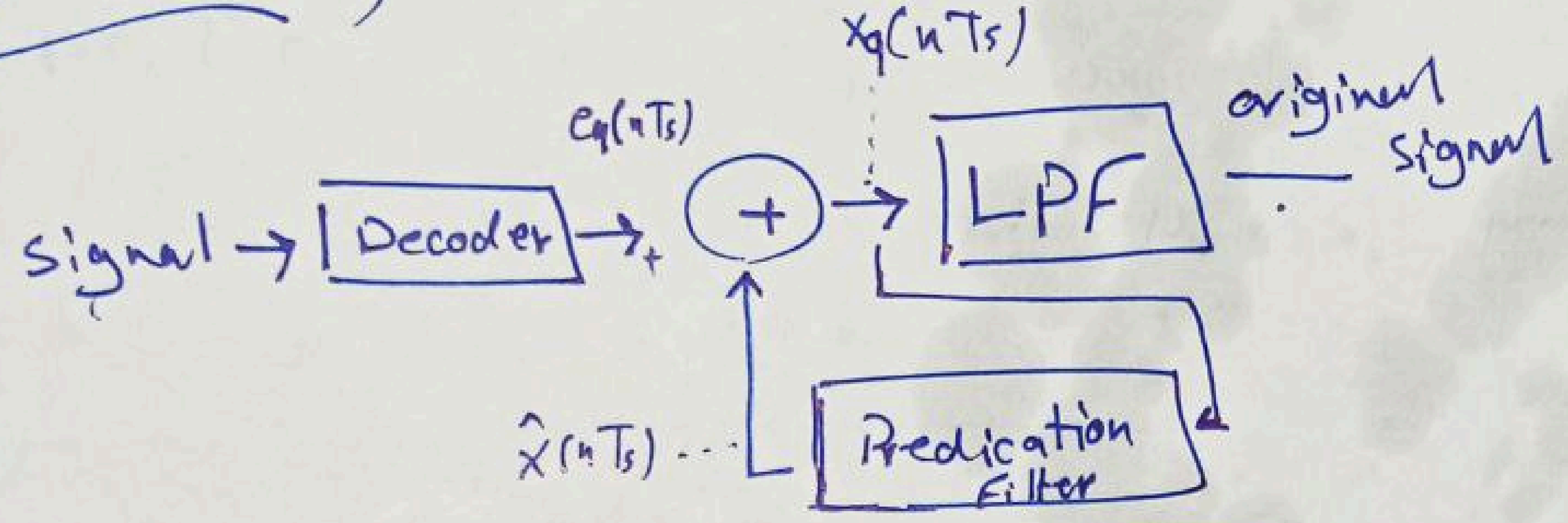
To maintained the same t. rate, (n) has to be incresed

$L = 2^n \qquad n = 8 \xrightarrow{\text{increased}} n = 10 \Rightarrow L = 2^{10} = 1024$

$$\text{SNR} = \frac{3L^2}{\ln} = 102308.752 = \boxed{50.1 \text{ dB}}$$

٢٠٢٥/٣/٢٠ تكملة السؤال

Decoder :



$e_q(n)$	$\hat{x} = x_q(n-1)$	$x_q = \hat{x}(n) + e_q(n)$
2	0	0+2 = 2
0	2	2+0 = 2
0	2	2+0 = 2
1	2	2+1 = 3
0	3	3+0 = 3
0	3	3+0 = 3

received signal = {2, 2, 2, 3, 3, 3}

* في حالة ذي ~~midriser~~ $\boxed{\text{mid riser}}$

راح يختلف الحل : مجرد عند التقريب نقرب ل

تاخذ بنظر الاعتبار 0.5 بين كل رقمين

* The function of (PAM) is Modifying the amplitude of pulses.

* The function of (PWM) is Modifying the width of pulse.

* The function of (PPM) is change the position of samples.

* (F_s) is the number of samples per second
  when :-
  1) [$F_s = 2B$] ⇒ The signal will be perfect.
  2) [$F_s > 2B$] ⇒ we use this case to improve reconstruction
  3) [$F_s < 2B$] ⇒ Aliasing occurs, There will be distortion

* What is aliasing ?

  It's the distortion by Lack of sample
  OR ⇒ by insufficient samping
  OR ⇒ by under sampling

* قد يطلب الميزة الاساسية التي تميز الـ (Digital) عن الـ (Analog)
وهي أن الـ (Digital) أفضل في التخلص من الـ (noise) و
الـ (distortion).

* The main function of aliasing Is to prevent aliasing by restricting the signal's band width.
بمعنى يتم تحديد (تقييد الاشارة) لعدم الانحياز خارج النطاق.

* تم ذكر بعض الملاحظات حول practical reconstruction Filters

مثلاً :- كيفية استخدامه والفائدة منه وأنواعه .

الاستخدام :- لإعادة بناء الإشارة

أنواعه :- Sharp cutoff

الأكثر إستخداماً لأنه ⇐ Gradual cutoff
أكثر فعالية ويستخدم لإزالة الترددات العالية غير المرغوبة

in this case ⇒ Sampling rate > Nyquist rate

↳ reorganize the Signal will be easier and use simpler Filters.

* The samples must be taken at equal intervals

* The role of aliasing Filter is :-

Prevent Spectral overlap of the Sampled signal.

* The main steps in (A/D) conversion is sampling.

* The main function of quantization in (A/D) is Approximat discrete analog values to the nearest digital Level.

* The main function of coding is to convert the quantized signal into binary form.

* The main function of Nyquist rate is Builds signals in an ORganized and Perfect way (reconstruction)

* The main function of [TDM] is transmit several signals at different intervals

Often we $\boxed{\text{can't}}$ use [LowPass_Filter], but we can [High OR Band Pass Filter]

في هذه الحالة قد يطلب السبب فهو يتعلق بال (Power)

قد تكون ال (Power) في حالة ال (Lowpass Filter) عالية جداً أو لا نهائية ففي مواضع معينة لا يمكن استخدامها.

لذلك استخدامات ال (Lowpass_Filter) تكون :

1) ~~After~~ Befor sampling.

2) and to recover the original signal.

3) in practical reconstruction Filters

Communication system II ٢٠٢٥/٤/١٧

~~A~~ Bipolar

0 → ~~No pulse~~ No pulse

1 → +ve or -ve

Problem: string "0"

Sol:- Adding puleses when no. (nomber) of "0" exceed S(W) ..

.. HDBN

High Density Bipolar signalling

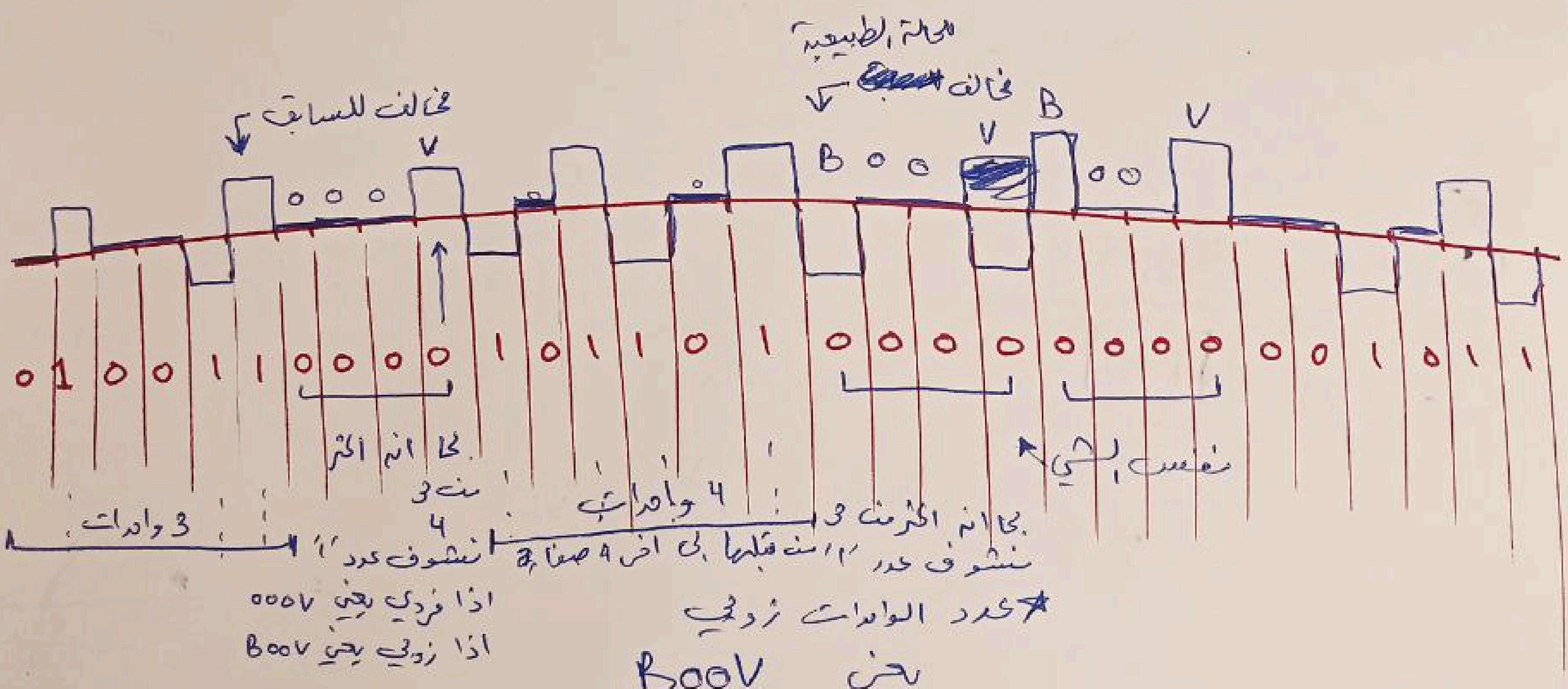
التقنية

HDB 3

عندما يكون عدد الواحدات قبل الاصفار هو عدد فردي] ⇒ 000V → violation (0, مخالف Bipolar)

عندما يكون عدد الواحدات قبل الاصفار هو عدد زوجي] ⇒ B00V → Bipolar

Ex مثال



7.2-1 full width $p(t) = \Pi(t/T_b)$

ⓐ PSD ? → Polar, on-off, bipolar signaling ?

ⓑ sketch PSD & B.W

ⓐ Given full width rectangular pulse $p(t) = \text{rect}\left[\frac{t}{T_b}\right]$

F.T from the table $P(w) = T_b \,\text{sinc}\left(\frac{wT_b}{2}\right) \dots ①$

① for Polar

PSD $S_y(w) = \dfrac{|P(w)|^2}{T_b} = \dfrac{T_b^{\cancel{2}} \,\text{sinc}^2\left(\frac{wT_b}{2}\right)}{\cancel{T_b}}$

∴ PSD is $T_b \,\text{sinc}^2\left(\frac{wT_b}{2}\right)$

② for on-off

$$S_y(w) = \frac{|P(w)|^2}{4T_b} * \left[1 + \frac{2\pi}{T_b} \sum_{-\infty}^{\infty} \delta\left(w - \frac{2\pi n}{T_b}\right)\right]$$

نعوض P(W) ونختصر

$$\Rightarrow S_y(w) = \frac{T_b}{4} \,\text{sinc}^2\left(\frac{wT_b}{2}\right)\left[1 + \frac{2\pi}{T_b} \sum \delta\left(w - \frac{2\pi n}{T_b}\right)\right]$$

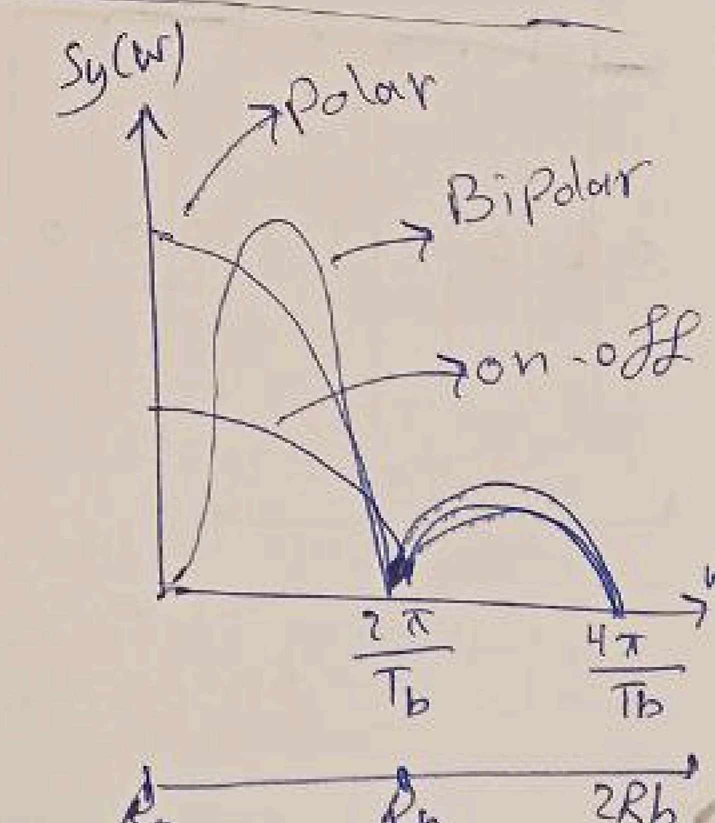
نختصر حسب خصائص on-off موجود بالمحاضرة

$$S_y(w) = \frac{T_b}{4} \,\text{sinc}^2\left(\frac{wT_b}{2}\right) + \frac{\pi}{2}\delta(w)$$

③ For Bipolar

$$S_y(w) = \frac{|P(w)|^2}{T_b}\left[1 - \cos(wT_b)\right]$$

$$= T_b \,\text{sinc}^2\left(\frac{wT_b}{2}\right) \sin^2\left(\frac{wT_b}{2}\right)$$



7.2-1 full width $p(t) = \Pi(t/T_b)$

(a) PSD ? → Polar, on-off, bipolar signaling ?

(b) sketch PSD & B.W

(a) Given full width rectangular pulse $p(t) = \text{rect}\left[\frac{t}{T_b}\right]$

F.T from the table $P(w) = T_b \text{ sinc}\left(\frac{wT_b}{2}\right) \cdots (1)$

(1) for Polar

PSD $S_y(w) = \frac{|P(w)|^2}{T_b} = \frac{T_b^{\cancel{2}} \text{sinc}^2\left(\frac{wT_b}{2}\right)}{\cancel{T_b}}$

$\therefore$ PSD is $T_b \text{sinc}^2\left(\frac{wT_b}{2}\right)$

(2) for on-off

$$S_y(w) = \frac{|P(w)|^2}{4T_b} * \left[1 + \frac{2\pi}{T_b} \sum_{-\infty}^{\infty} \delta\left(w - \frac{2\pi n}{T_b}\right)\right]$$

تعوض P(W) وتختصر

$$\Rightarrow S_y(w) = \frac{T_b}{4} \text{sinc}^2\left(\frac{wT_b}{2}\right)\left[1 + \frac{2\pi}{T_b} \sum \delta\left(w - \frac{2\pi n}{T_b}\right)\right]$$

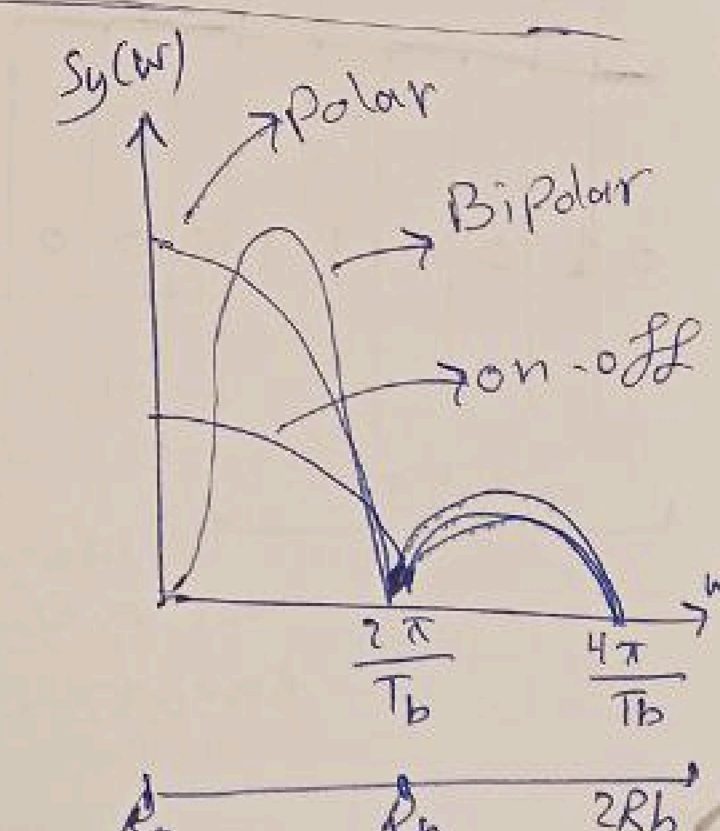
نتذكر من خواص on-off موجود بالمحاضرة

$$S_y(w) = \frac{T_b}{4} \text{sinc}^2\left(\frac{wT_b}{2}\right) + \frac{\pi}{2}\delta(w)$$

(3) For Bipolar

$$S_y(w) = \frac{|P(w)|^2}{T_b}\left[1 - \cos(wT_b)\right]$$

$$= T_b \text{ sinc}^2\left(\frac{wT_b}{2}\right) \sin^2\left(\frac{wT_b}{2}\right)$$



lathi ملاحظة   22/2/2024   Communication Tutorial   First Lec.

6.2.1: (a)   ملاحظة $L = 2^n$

since Number of characters = 128 $\Rightarrow 128 = 2^7$

$\therefore$ Number of bits required per character = 7 bits/character.

(b) Computer generates 100,000 character/second.

$\therefore$ Number of bits/second required to transmit this output

$= 7 \times 100{,}000$ bits/s $= 700{,}000$ bits/s $= 700k$ bits/s

$\therefore$ Minimum bandwidth required to transmit this signal

$= \frac{7 \times 100{,}000}{2} = 350{,}000 = 350\,KHz$ $\therefore$ Min. B.W $= 350\,KHz$

(c) An additional parity bit is added.

$\therefore$ Number of bits required per character $= (7+1)$ bits/char.

$= 8$ bits/char.

And No. of bits/sec. required transmit the o/p

$= 8 \times 100{,}000 = 800{,}000 = 800\,kbits/s.$

$\therefore$ Min. B.W. required $= \frac{8 \times 100{,}000}{2} = 400{,}000 = 400\,KHz$

6.2.2: (a) $\because$ B.W is 15 KHz $\therefore$ The Nyquist rate = 30 KHz

(b) $L = 2^n \Rightarrow 65536 = 2^n \Rightarrow \therefore n = 16$

$\therefore$ so that 16 binary digits are needed to encoded each sample.

(c) $30{,}000 \times 16 = 480000$ bits/s $= 480k$ bits/s

(d) $44{,}100 \times 16 = 705600$ bits/s

6.2.3: (a) Given B.W = 4.5 MHz $\therefore$ Nyquist rate

$= 2 \times 4.5\,MHz$

$= 9\,MHz$

sampling rate is 20% above Nyquist rate

$\therefore$ sampling rate $= \left[9 + \frac{20}{100} \times 9\right] = [9 + 1.8] = 10.8\,MHz$

(b) $L = 2^n \Rightarrow 1024 = 2^{10} \Rightarrow$ so that 10 bits or binary pulses are neded to encode each sample.

(c) $10.8 \times 10^6 \times 10 = 108 \times 10^6 = 108$ M bit/s

[illegible] B.W

6.2.4: Each signal has $B.W = 1\,KHz$ ~~Nyquist rate ≤ 2kHz~~

∵ mp be the peak signal amplitude

∴ Quantization error $\leq \frac{0.2}{100} \times mp \times$ ~~Nyquist rate~~

$\cancel{\leq \frac{0.2}{100} \times \frac{10 \times mp}{2 \times 1000}}$

$\leq \frac{mp}{500}$ ... (1)

∵ max. quantization error $= \frac{\Delta V}{2} = \frac{2mp}{2L} = \frac{mp}{L}$

where L = No. of quantization Levels

$\therefore \frac{mp}{L} = \frac{mp}{500} \Rightarrow \therefore L = 500$

Because ~~power~~ L should a power of 2 we chose ~~L=512~~

$L = 512 = 2^9 \Rightarrow$ This requires a 9. binary code per sample

Nyquist rate $= 2 \times B.W = 2\,KHz$

$\left[2000 + \frac{20}{100} \times 2000\right] = [2000 + 400] = 2400\,Hz$

Thus each signal has 2400 sample/sec and each sample is encoded by 9 bit ∴ each signal uses $9 \times 2400 = 21.6$ k b/s

∵ 5 signals are multiplexed ⇒ we need a total $5 \times 21.6k = 108$ k b/s dat bits

Framing and synchro. required aditional 0.5% bits

$108000 \times \frac{0.5}{100} = 540$ bits ⇒ ∴ The total $108k + 540 = 108540$ b/s

Then the min. B.W is $\frac{108.54}{2} = 54.27$ KHZ

6.2.6 Because sinsiodal signal then

$$V_{r.m.s} = \frac{V_p}{\sqrt{2}} \Rightarrow \frac{\widetilde{m^2(t)}}{m_p^2} = \frac{\left(\frac{m_p}{\sqrt{2}}\right)^2}{m_p^2} = \frac{\frac{m_p^2}{2}}{m_p^2} = 0.5$$

$\because$ ~~F~~message transmit without Compression

$\frac{\widetilde{m^2(t)}}{m_p^2}$ where m_p peak signal

$\widetilde{m^2(t)}$ is mean Squar value of $m(t)$

$$\because SNR = 47\,dB \Rightarrow \frac{S_o}{N_o} = 3L^2 \frac{\widetilde{m^2(t)}}{m_p^2}$$

in Linear $47 = 10\log_{10}(X) \Rightarrow 4.7 \log_{10}(X) \Rightarrow SNR = 10^{4.7}$

$= 50118.7$

≈ 50119

$$3L^2 * (0.5) = 50119$$

$$L^2 = \frac{50119}{1.5} = 33413 \Rightarrow \therefore L = 182.8$$

$$n = \lceil \log_2(L) \rceil = \log_2(182.8) = 7.5 \approx 8$$

$$L = 2^8 \Rightarrow 256$$

$$\therefore SNR = 3L^2 \frac{\widetilde{m^2(t)}}{m_p^2} = 3 * (256)^2 * 0.5$$

$$= 65536 * 1.5 = 98304$$

$$\text{in dB} = 10\log_{10}(98304) = 4.992 * 10 = 49.92\,dB$$

NOTE 40 Pro 24mm f/1.75 1/50s ISO311

6.2.5: $\because B.W = 100\,Hz \quad \therefore$ Nyquist rate $= 2 * B$
$= 2 * 100 = 200\,Hz$

Then the sampling rate $f_s = 2 *$ Nyquist rate $= 2 * 200 = 400\,Hz$ for 1 signal

I have 10 patient = 10 samples

For (10 paient) $= 10 * 400 = 4000$ sample/sec.

~~Quantization error $\leq \frac{0.25}{100} * mp$~~

~~$\frac{mp}{L}$ $\Rightarrow$ Quantization error $\leq \frac{0.25}{100} * mp$~~

~~quan error~~

$\therefore$ the Δv is quantization step عرض

$T_s = \frac{1}{400} \Leftarrow T_s = \frac{1}{f_s}$ sample interval تتناسب مع

$\frac{\Delta v}{} = \frac{2mp}{L} \Rightarrow \frac{\Delta v}{2} = \frac{mp}{L} =$ quantization error

Quantization error $\leq \frac{0.25}{100} * mp \leq \frac{mp}{400}$

$\frac{mp}{L} = \frac{mp}{400} \Rightarrow \therefore L = 400$

لازم عشان أقرب ح n لـ L

$L = 2^n \Rightarrow 400 = 2^n \Rightarrow n = \lceil \log_2(L) \rceil$

$\log_2(400) = 8.6 \simeq 9 \quad \therefore L = 2^9 = 512$

$\therefore 9$ bit/sample $\Rightarrow$ Min. bitrate $= 9 * 4000$
$= 36k$ bi/sec.

~~$\therefore$ Min B.W~~ $=$ Nyquist rate $= 2 * B$

$\therefore$ Min. B.W $= \frac{36k}{2} = 18k$ b/sec.

ليش قسمة على (2) B.W لأن من

6.2-6, مثال